

# Grok 4.5 Under the Hood

## Token Efficiency vs. Trust

*The launch that split into two separate stories — a genuine cost-math breakthrough for agentic work, and an unresolved fight over who controls what the model is allowed to say*

Grok 4.5 shipped July 8 alongside GPT-5.6 and Claude Cowork's mobile expansion — Day 105 covered the surface war between the labs. Today we go under the hood on xAI's model specifically, because the launch produced two genuinely separate stories that both matter if you're deciding whether to route work to it. The first is a real efficiency number: on SWE-Bench Pro coding tasks, xAI reports Grok 4.5 used an average of 15,954 output tokens per task against 67,020 for Claude Opus 4.8 — a 4.2x gap — while pricing input tokens at \$2 per million. The second is messier: an independently measured hallucination rate that more than doubled from the prior version, and a political-bias debate that became the single loudest thread on the model's own Hacker News launch post. 105 days into this series we've tracked the frontier race compress to a near-tie (Day 80) and watched three labs ship within the same 48 hours (Day 105) — today is about what happens when you look past the headline benchmark number at the model that actually shipped.

|                                                                                    |                                                              |                                                               |                                                                  |
|------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------|
| <b>4.2x</b><br>Token efficiency vs. Opus 4.8<br>on the same SWE-Bench Pro<br>tasks | <b>\$2</b><br>per million input tokens —<br>xAI's list price | <b>54%</b><br>Hallucination rate, up from<br>~25% on Grok 4.3 | <b>#4</b><br>Rank on Artificial Analysis's<br>Intelligence Index |
|------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------|

### 1) The token-efficiency claim that actually changes agentic cost math

"Token efficiency" sounds like a footnote until you see what it does to agentic economics. xAI reports Grok 4.5 completed SWE-Bench Pro coding tasks using an average of 15,954 output tokens, versus 67,020 for Claude Opus 4.8 on the same benchmark — a 4.2x difference — while pricing input tokens at \$2 per million. For a single chat question, the per-token list price is what matters. For an agentic loop that might call a model hundreds of times to finish one task, the number of tokens burned per step compounds directly into dollars: a model that's only slightly cheaper per token but needs 4x as many tokens to finish the same job ends up more expensive in practice. That's the gap most leaderboards hide — they report accuracy per task, not tokens spent getting there, which is exactly the number that decides your bill at agentic scale.

| Model           | Avg. output tokens (SWE-Bench Pro) | Input price / 1M tokens |
|-----------------|------------------------------------|-------------------------|
| Grok 4.5        | 15,954                             | \$2.00                  |
| Claude Opus 4.8 | 67,020                             | Premium-tier pricing    |
| Efficiency gap  | 4.2x fewer tokens for Grok 4.5     | —                       |

## 2) The hallucination spike the cost math doesn't show

The same launch that produced the efficiency number also produced a reliability warning: independent evaluators measured Grok 4.5's hallucination rate at roughly 54%, more than double the ~25% recorded for Grok 4.3. That matters because token-efficiency savings evaporate fast if every other output needs a human to catch a fabricated fact — and in an agentic pipeline, one hallucinated intermediate step can cascade into a confidently wrong final answer with nobody watching until it's shipped. Cheap and fast doesn't help if you have to bolt on a verification layer to trust the result, which is its own token — and time — cost that the efficiency benchmark never counts.

| Version  | Hallucination rate | What changed                            |
|----------|--------------------|-----------------------------------------|
| Grok 4.3 | ~25%               | Prior-generation baseline               |
| Grok 4.5 | ~54%               | More than double, per independent evals |

## 3) The political-bias debate: unresolved, not settled

On Hacker News, the loudest single thread on Grok 4.5's launch post wasn't about capability at all — it was about trust, specifically whether Elon Musk's team nudged the model's outputs on political questions. The evidence cited included a reported system-prompt instruction telling Grok to be "politically incorrect," which researchers argue may have overcorrected the model into taking contrarian positions across the board rather than one specific lean. Independent bias testing found a genuinely mixed picture: Grok skews right of most other assistants but still left of center overall, and is, oddly, harsher on Musk's own companies than any other AI tested — the opposite of the "biased in the owner's favor" accusation. The debate stayed unresolved in the comments: one tester reported Grok as "more politically correct than GPT and Gemini" in daily use, directly contradicting the "nudged" narrative in the same thread. The real takeaway isn't that Grok is proven biased or proven clean — it's that shipping a system-prompt instruction like "be politically incorrect" turns every output into something users now have to second-guess, whichever direction it actually pushes.

| When to reach for it                                      | Use                                  | Guardrail                                                               |
|-----------------------------------------------------------|--------------------------------------|-------------------------------------------------------------------------|
| Need the cheapest cost on a long agentic loop             | Grok 4.5                             | Budget a human-review pass for the higher hallucination rate            |
| Need output you can trust without double-checking         | A verified pipeline around any model | Don't skip fact-checking just because the eval score is high            |
| Evaluating a model for consequential or political content | Ask for the system prompt            | A hidden steering instruction can matter more than the leaderboard rank |

**THE ONE RULE**

A benchmark score tells you what a model can do under ideal conditions; it doesn't tell you what it will do at 2am inside your agent loop with nobody watching. Grok 4.5's token-efficiency number is real and worth using — but pair it with your own hallucination check, and ask any lab for its system prompt before you trust its judgment on anything that matters.

**VIRAL / OPEN-SOURCE SPOTLIGHT**

**Kling AI's photo-to-dance trend turns any picture into a viral dance video**

Kling AI's motion-control feature works like digital puppetry: upload a single photo, pick from 5,000+ dance and motion templates (or upload your own reference clip), and the model transfers the movement, timing, and rhythm from the reference onto the still photo while preserving the subject's identity — turning a static picture into a few seconds of full-body dance footage generated in roughly the time it takes to read this sentence. It's spread fastest through "baby dance" clips, where a photo of an infant is animated into a K-pop or shuffle routine, racking up millions of views on TikTok and Instagram Reels. It's viral for the same reason every low-effort, high-output AI trend goes viral: the barrier to making something shareable dropped to "have one photo," and the output looks polished enough to pass as real motion capture to a casual scroller — no camera, choreography, or editing skill required.

|                                                       |                                                  |                                            |                                                       |
|-------------------------------------------------------|--------------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| <b>5,000+</b><br>Dance and motion templates available | <b>1</b><br>Photo needed to generate a full clip | <b>Seconds</b><br>Generation time per clip | <b>Millions</b><br>of views on baby-dance clips alone |
|-------------------------------------------------------|--------------------------------------------------|--------------------------------------------|-------------------------------------------------------|

## MARKET SIGNAL

Grok 4.5's launch is the clearest evidence yet that "smartest model" and "model you should route work to" are no longer the same question. xAI is winning the pure cost-per-completed-task race — 4.2x fewer tokens for the same coding benchmark is a real number engineering teams will act on. But the same week that number shipped, its own hallucination rate doubled and its own launch thread argued about whether its answers can be trusted at all. Day 105 called this the shift from "whose model is smartest" to "whose agent do you trust to work unsupervised" — Grok 4.5 shows that trust question now splits into two separate axes: do you trust its facts, and do you trust its neutrality. A lab can win the efficiency number and still lose the adoption decision if either one stays unresolved.

## PRACTICAL TAKEAWAYS

### 1. Price agentic workloads per completed task, not per token

Compare total tokens burned across an entire agentic run, not the sticker price per million tokens — Grok 4.5's 4.2x efficiency gap on SWE-Bench Pro would be invisible if you only compared list prices.

### 2. Pair any efficiency win with your own hallucination check before trusting it

A rate that jumps from ~25% to ~54% between versions means the previous version's reliability numbers tell you nothing about the current one — re-test before routing production work to a new release, however good its cost math looks.

### 3. Ask for the system prompt before you ask for the benchmark score

The "politically incorrect" instruction reportedly baked into Grok's system prompt mattered more to how people evaluated the model than any leaderboard rank — for any model handling consequential or subjective content, request the operator's steering instructions, not just its eval results.

## TOMORROW

Tomorrow — Day 107: Formal Verification Meets LLMs — Mistral's Leanstral 1.5 proves code correct with mathematical certainty via Lean 4 instead of just testing it, and what that means for how much you can actually trust AI-written software.